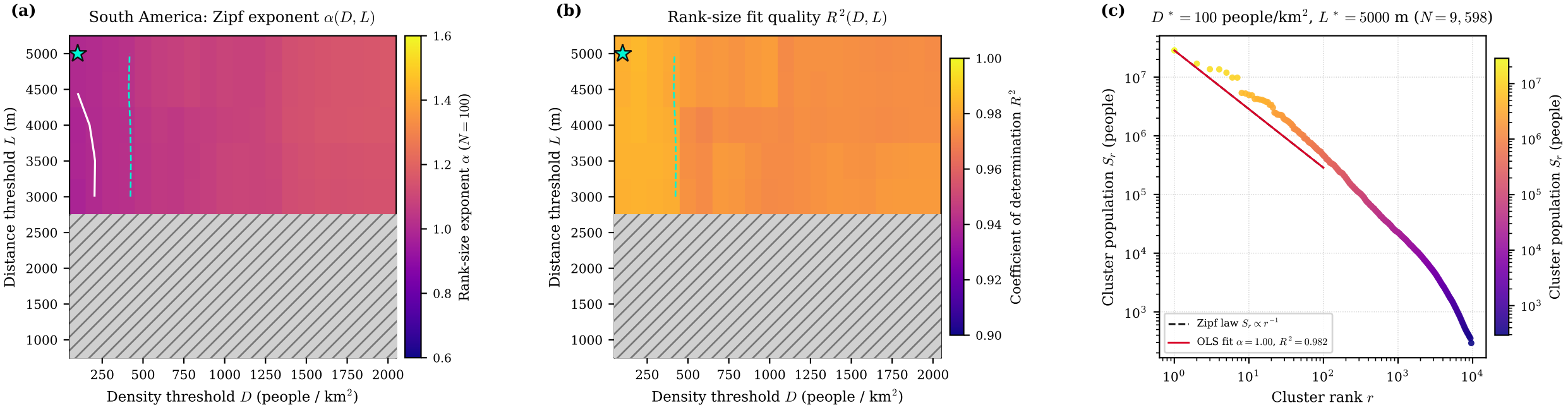

(a)
South America: Zipf exponent $\alpha(D, L)$
Distance threshold L (m)
Density threshold D (people / km^2)
Rank-size exponent α ($N = 100$)
(b)
Rank-size fit quality $R^2(D, L)$
Distance threshold L (m)
Density threshold D (people / km^2)
Coefficient of determination R^2
(c)
$D^* = 100$ people/km^2, $L^* = 5000$ m ($N = 9,598$)
Cluster population S_r (people)
Cluster rank r
Zipf law $S_r \propto r^{-1}$
OLS fit $\alpha = 1.00$, $R^2 = 0.982$